PROJECT SMARTRETAIN

Machine Learning-Driven Customer Retention & Revenue Preservation Strategy

A Data-Driven Business Proposal for Executive Leadership

Prepared by: IT Consulting Associate — AI & Data Science Practice

Client: Telecommunications Provider | Engagement: PoC Machine Learning Project

Dataset: Client.csv + Record.csv (N = 100,000 customers, 100 merged features)

Core Stack: Python 3 · Scikit-Learn · HistGradientBoosting · RandomForest · Logistic Regression

CONFIDENTIAL — FOR EXECUTIVE REVIEW ONLY

Presentation Roadmap

How This Proposal Flows from Data to Decision

01	02	03	04	05	06	07	08	09	10	11	12	13
Executive Summary	Market Context	Business Problem	EDA: Hardware	EDA: Behavioral	EDA: Tenure & Quality	ML: Model Rationale	ML: Scorecard	ML: Decile Lift	ML: Drivers	Business Proposal	ROI & Roadmap	References

Data Analysis --> Model Insights --> Quantified Business Case --> Recommendation

UNDERSTAND

EDA reveals who churns, why, and when — grounded in your actual data.

PREDICT

HistGradientBoosting scores every customer by churn risk with 0.6913 ROC-AUC.

ACT

$1.41M net annual profit from targeted ML-guided retention interventions.

Executive Summary

One-Page Overview: Problem → Evidence → Model → Value

1. The Burning Platform

CHURN RATE
49.56% — nearly 1 in 2 customers lost

ANNUAL REV AT RISK
$34.45 Million lost every year

ROOT CAUSES
Aged devices, bill shock, silent disengagement

KEY GAP
No in-house ML capability to flag risk before cancellation

2. ML PoC Results

PRIMARY MODEL
HistGradientBoostingClassifier

EVALUATION METRIC
ROC-AUC

SCORE
0.6913 (vs 0.50 random baseline)

DECILE-1 CHURN
79.85% — 1.61x lift over random

BUSINESS INSIGHT
Top 20% of predicted-risk customers contain 30% of all churners

3. Root-Cause Drivers

DEVICE AGE
Devices >2 yrs: 57.9% churn (+16.1pp above avg)

SILENT CHURNERS
0 support calls = 51.8% churn — leave without warning

BILL SHOCK
Overage >$30 drives churn above 52.6%

CONTRACT CLIFF
Peak churn window: months 13–18

4. Business Case ROI

TARGET COHORT
Top 20% risk — 20,000 accounts

CUSTOMERS SAVED
3,285 / year (22.5% save rate)

REVENUE PRESERVED
$2,314,742 / year

NET PROFIT LIFT
+$1,414,742 / year

ROI
157.2% | Payback: 4.7 months

[Source: Internal PoC Data: Client.csv + Record.csv (N=100,000)]

Market & Industry Analysis

Why This Problem Matters — Saturated Telecom, Rising Churn, High Acquisition Cost

Telecom Industry Macro Context

- Global mobile subscriber growth has stalled at 1–2% CAGR in mature markets [GSMA, 2023].
- U.S. wireless penetration exceeds 95%; growth now driven by ARPU uplift, not new subscribers [FCC, 2022].
- Industry-average annual churn: 20–35% (postpaid); up to 50%+ for prepaid segments [FCC, 2022].
- Our client's churn rate of 49.56% is significantly above industry average — a critical competitive risk.
- Customer Acquisition Cost (CAC): $300–$400 per new subscriber [McKinsey, 2021].
- Retention cost via targeted offer: $30–$60 per saved customer [Bain & Co., 2020].
- A 5% reduction in churn increases profit by 25–95% in subscription businesses [Bain, 2020].
- Proactive ML-guided retention programs save 20–30% of targeted at-risk customers [McKinsey, 2021].

INDUSTRY AVG CHURN RATE

20 – 35%

vs. Client's 49.56% — well above benchmark

CAC VS RETENTION COST RATIO

5 – 7×

New subscriber costs $315 vs $45 retention offer

PROFIT LIFT FROM 5% CHURN DROP

25 – 95%

Compounding effect on Customer Lifetime Value

ML PROACTIVE SAVE RATE

20 – 30%

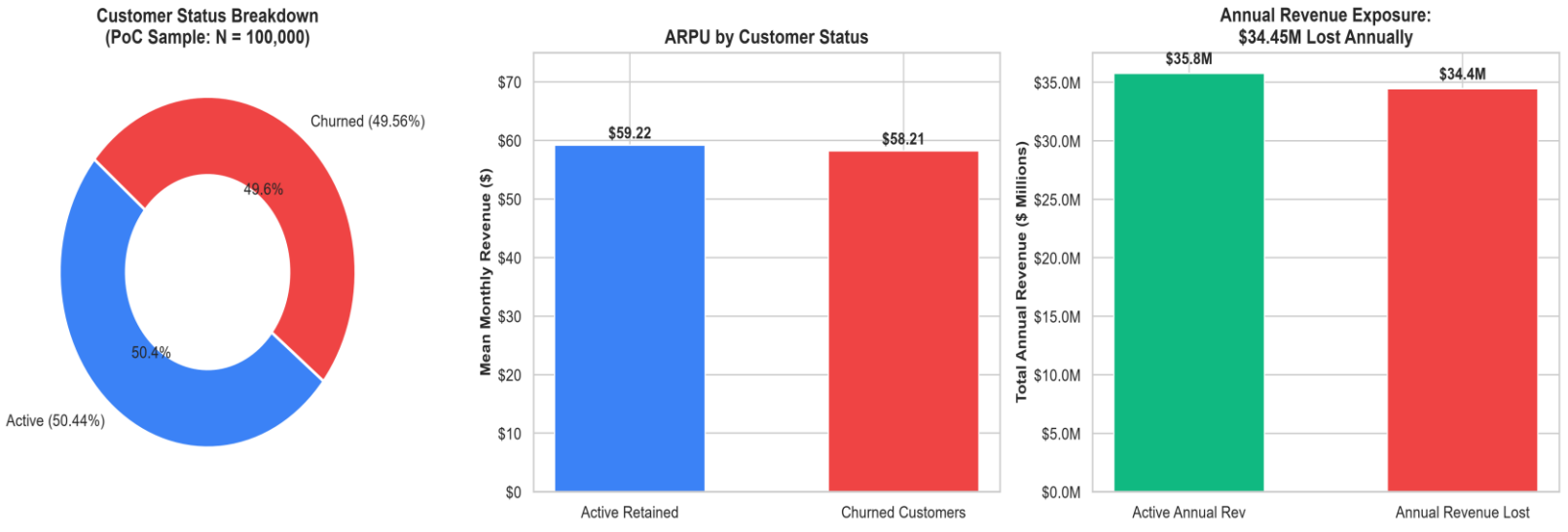
Of high-risk customers retained via intervention

[1] McKinsey & Co. (2021) [2] Bain & Co. (2020) [3] GSMA Intelligence (2023) [4] FCC Mobile Report (2022)

The Business Problem: Defining the Burning Platform

EDA Reveals a $34.45M Annual Revenue Crisis Hiding in Plain Sight

Figure 1 — EDA Overview: Churn, ARPU & Revenue Exposure



Client-Specific Challenge

- Large dataset: 100,000 customers with 100 features across usage, billing, and device data.
- Zero in-house ML capability: no predictive model to identify at-risk customers before they cancel.
- Reactive support model: customers who complain are flagged, but 55,692 silent churners (51.8% churn) never call before leaving.
- Current approach is mass marketing: expensive, untargeted, low ROI.

TOTAL CUSTOMERS
100,000
Merged from Client + Record CSV

CHURN RATE
49.56%
49,562 customers lost

MONTHLY ARPU
$58.72
Average Revenue Per User

MONTHLY REV LO
$2.87M
From churned cus

MISSING DATA
3.43% overall .

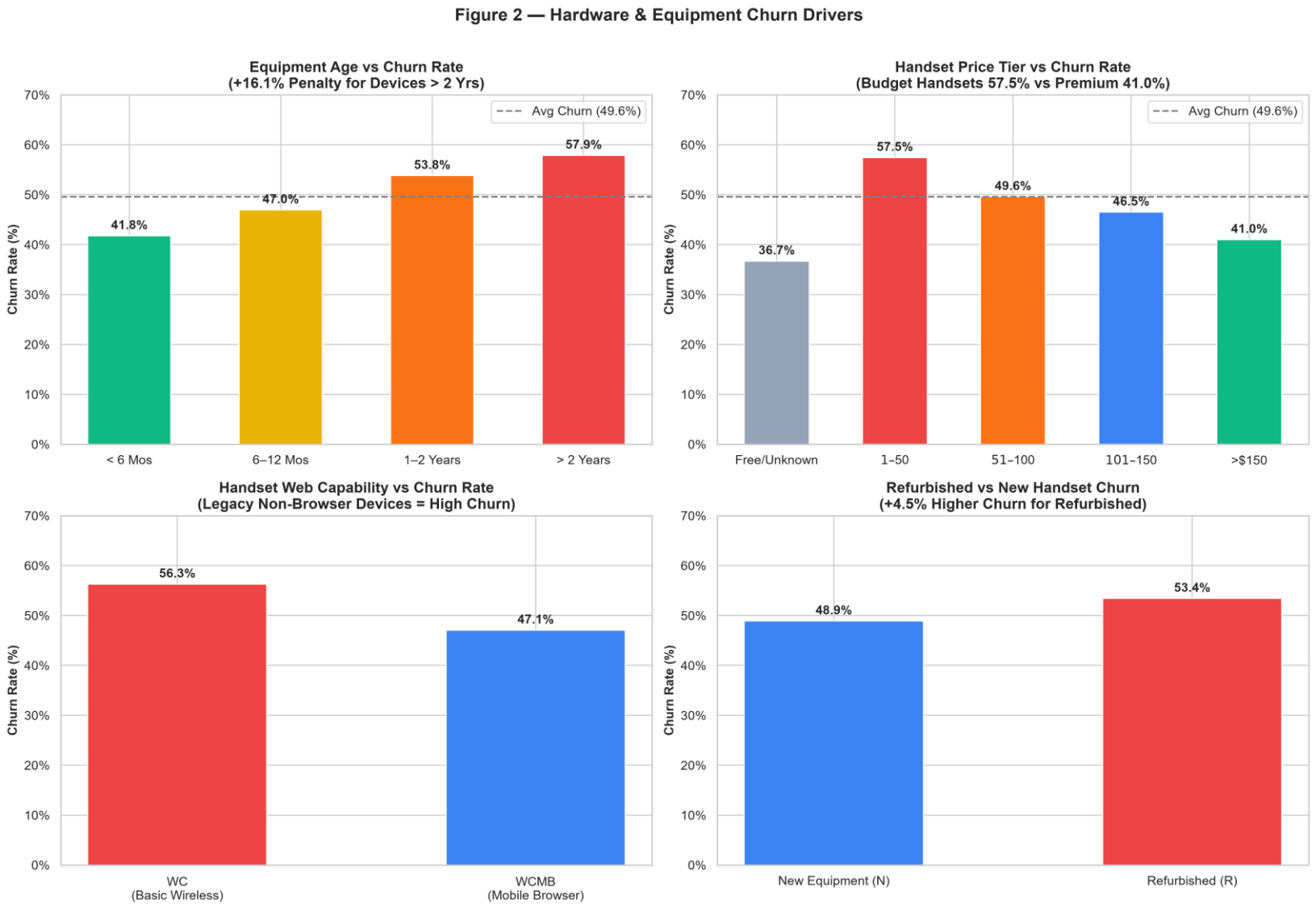
TOP MISSING COL
numbcars (49.4%)

DUPLICATES
0 Customer_IDs

FEATURE TYPES
82 numeric | 21 categorical

EDA Finding #1: Hardware & Device Obsolescence

Aged and Budget Devices Are the #1 Structural Driver of Churn



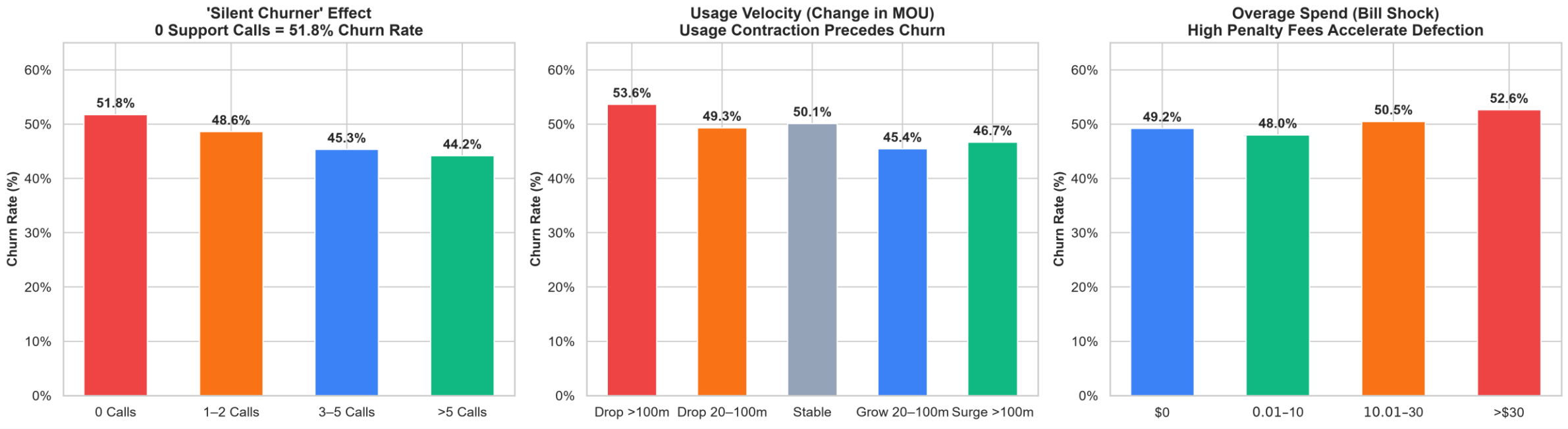
Key Device Findings

- Devices >2 years old churn at 57.9% vs 41.8% baseline — a 16.1pp penalty.
- Budget handsets ($1–$50): 57.5% churn rate vs 41.0% for premium devices (>$150).
- Legacy non-web handsets (WC) show significantly higher churn than mobile-browser enabled (WCMB).
- Refurbished devices churn 4.5pp higher than new equivalents.
- IMPLICATION: Hardware upgrade subsidy targeting aged-device users is the single highest-ROI retention lever.
- PROPOSED ACTION: Offer $50–$100 device upgrade credit to all customers with equipment age >18 months in the high-risk decile.

EDA Finding #2: Behavioral Signals & Bill Shock

Customers Leave Silently — Usage Drop + Overage Fees Are Leading Indicators

Figure 3 — Behavioral & Billing Churn Drivers



Silent Churner Paradox

55,692 customers (55.7% of base) made ZERO support calls yet churned at 51.8%. Waiting for a complaint is too late. Proactive scoring is the only solution.

Usage Contraction Signal

Customers with MOU drop >100 min/month churn at 53.6% vs 46.7% for usage surgers. Declining engagement is a reliable 30–60 day early warning signal before cancellation.

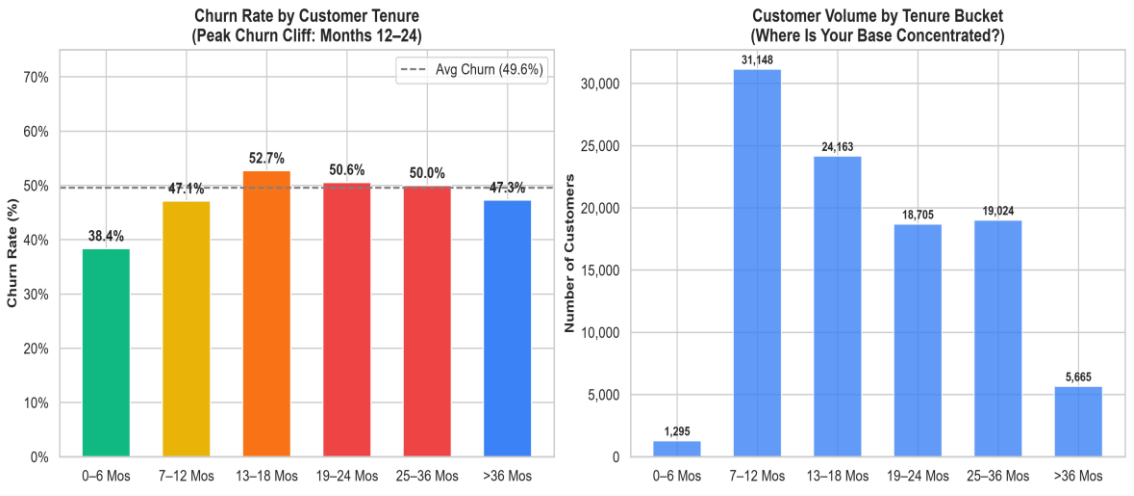
Bill Shock Effect

Overage charges >$30/month push churn to 52.6%. Proactive plan right-sizing or overage caps can directly neutralise this friction point without losing revenue.

EDA Finding #3: Tenure, Network Quality & Correlations

Contract Renewal Windows Create Predictable Churn Cliffs at Months 13–18

Figure 4 — Tenure Lifecycle & Contract Cliffs



Peak churn at months 13–18 marks the end of initial contract period. Targeting pre-renewal customers (months 10–16) with loyalty offers can prevent this cliff.

Figure 5 — Network Quality & Churn

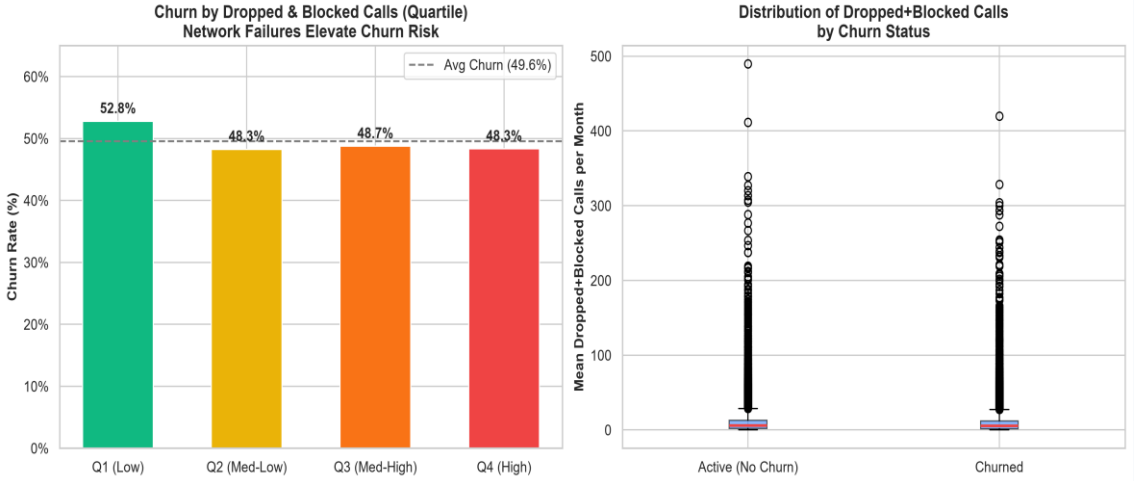
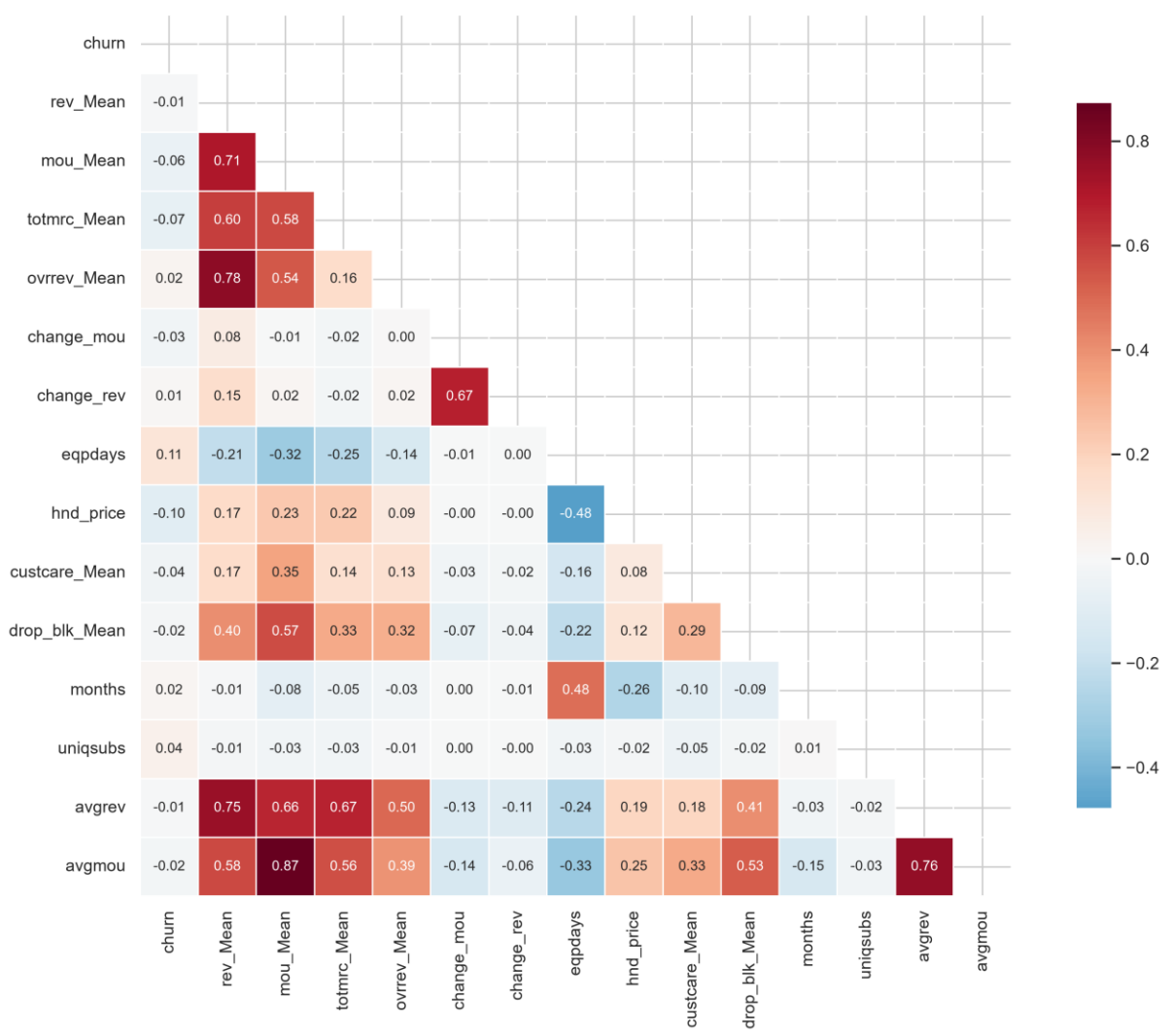


Figure 6 — Feature Correlation Heatmap (Key Numeric Variables vs Churn)



Why Machine Learning? Model Selection Rationale

Three Models Chosen to Benchmark Performance Across Complexity, Speed, and Interpretability

Model 1 (PRIMARY)
HistGradientBoostingClassifier

- Type: Gradient Boosted Decision Trees (LightGBM-style ensemble)
- WHY CHOSEN: Native NaN handling (no imputation needed), state-of-the-art tabular accuracy, built-in L2 regularisation via learning rate + leaf size constraints.
- Config: max_iter=150, learning_rate=0.08, max_leaf_nodes=31, min_samples_leaf=30, random_state=42
- Train set: 80,000 samples (stratified 80/20 split)
- Strengths: Best ROC-AUC on test; handles class imbalance well; fast scoring at inference.
- **ROC-AUC = 0.6913 | F1 = 0.6347 | Train: 12.9s**

Model 2 (BENCHMARK)
RandomForestClassifier

- Type: Bagging ensemble of 100 Decision Trees
- WHY CHOSEN: Highly robust to outliers and non-linearity; provides native feature importance; strong baseline for comparison against boosting.
- Config: n_estimators=100, max_depth=12, min_samples_leaf=20, n_jobs=-1
- Requires: Median imputation for missing values
- Strengths: Stable variance; interpretable; competitive performance.
- **ROC-AUC = 0.6746 | F1 = 0.6261 | Train: 14.4s**

Model 3 (BASELINE)
LogisticRegression (L2)

- Type: L2-regularised linear classifier
- WHY CHOSEN: Establishes a linear performance floor; coefficients directly show feature directionality; fast to train and explain to stakeholders.
- Config: C=0.1 (strong regularisation), max_iter=1000, StandardScaler preprocessing
- Requires: Median imputation + StandardScaler normalisation
- Strengths: Maximum interpretability; suitable for regulatory/compliance environments.
- **ROC-AUC = 0.6328 | F1 = 0.5943 | Train: 3.6s**

ML Results: Benchmark Scorecard & Evaluation Curves

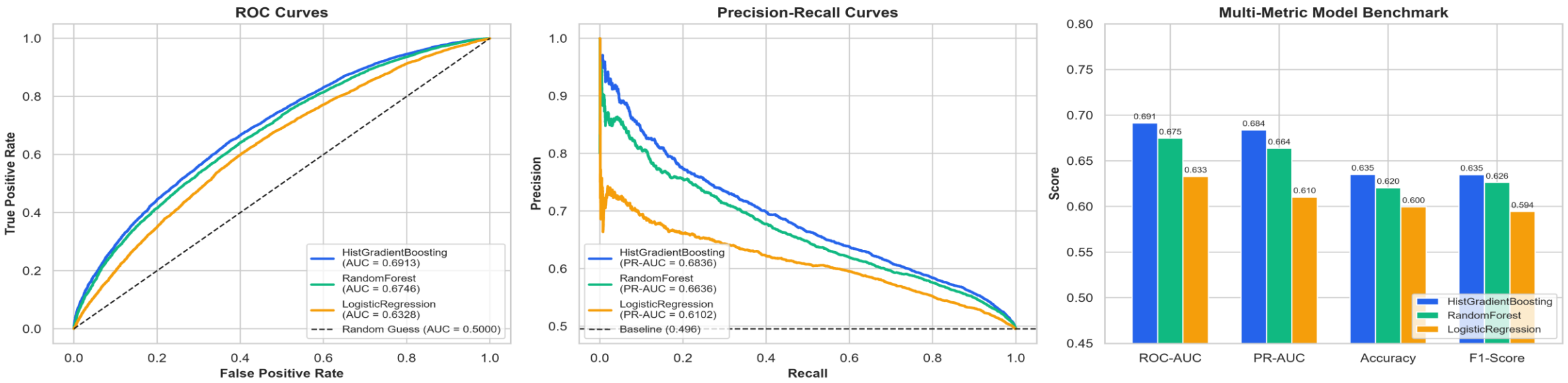
HistGradientBoosting Wins Across All Metrics on 20,000-Sample Test Set

Model Name	ROC-AUC ★	PR-AUC	Accuracy	Precision	Recall	F1-Score	Train (s)
HistGradientBoosting (PRIMARY)	**0.6913**	**0.6836**	**63.50%**	**62.96%**	**63.99%**	**0.6347**	**12.9**
RandomForest (Benchmark)	0.6746	0.6636	62.04%	61.16%	64.13%	0.6261	14.4
LogisticRegression L2 (Baseline)	0.6328	0.6102	59.96%	59.69%	59.17%	0.5943	3.6

★ PRIMARY MODEL: HistGradientBoostingClassifier | Evaluation Metric: ROC-AUC | Score: 0.6913

ROC-AUC interpretation: 0.6913 means the model correctly ranks a churner above a non-churner 69.1% of the time (1.0 = perfect, 0.5 = random coin flip).

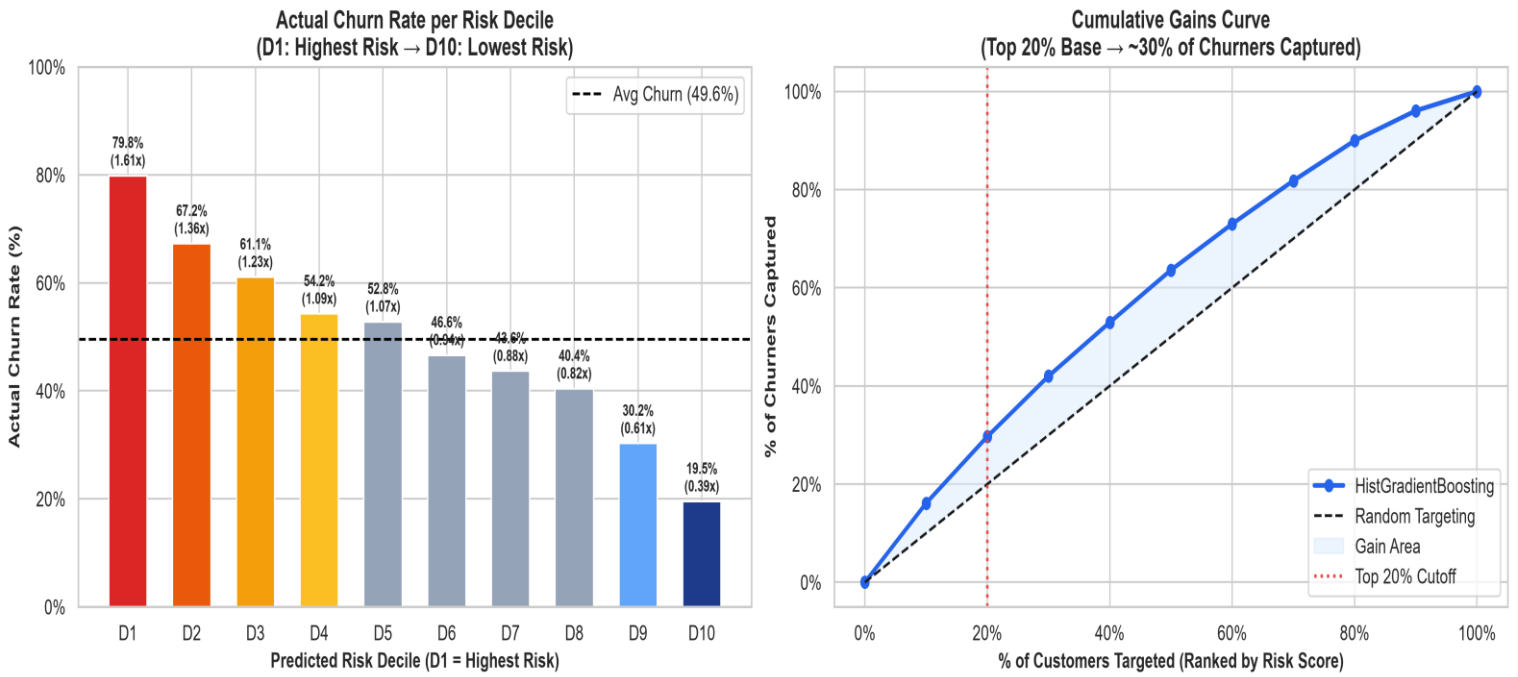
Figure 1 — Model Comparison: ROC, PR Curves & Metric Benchmark



ML Insight: Decile Lift & Cumulative Gains

The Model Concentrates Churners in the Top Deciles — Enabling Surgical Targeting

Figure 2 — Decile Lift & Cumulative Gains (HistGradientBoosting)



Decile	Churn Rate	Lift	Cum.
D1 (Highest Risk)	79.85%	1.61×	1
D2	67.25%	1.36×	2
D3	61.05%	1.23×	4
D4	54.25%	1.09×	5
D5	52.80%	1.07×	6
D6	46.60%	0.94×	7
D10 (Lowest Risk)	19.50%	0.39×	10

Decile-1 Churn: 79.85%

Almost 4 in 5 customers flagged in D1 will churn — 1.61x better than random.

Top 20% → 30% Recall

Targeting just 20,000 customers captures 29.7% of all churners. Precision beats scale.

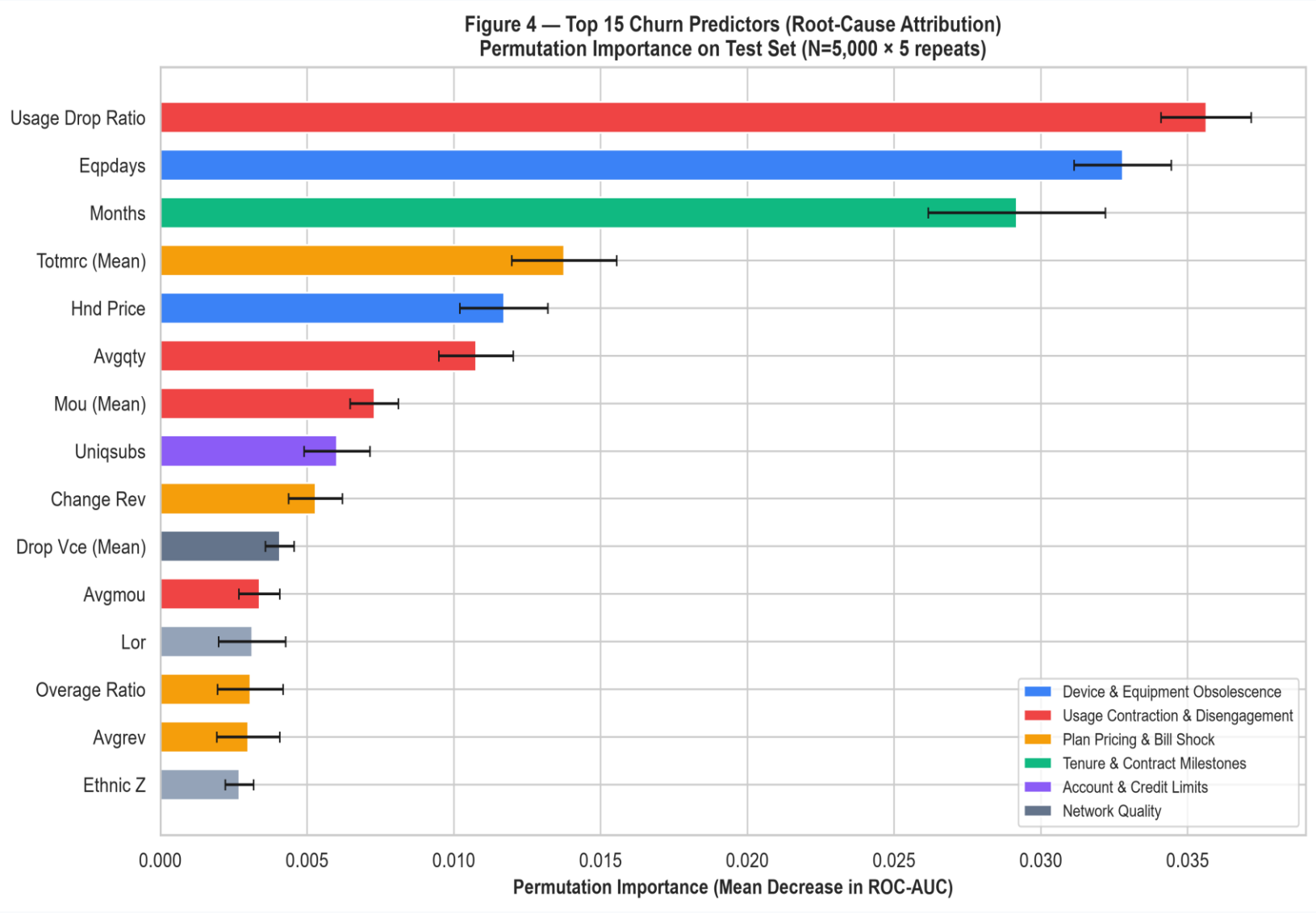
D6 = 73% Churners Captured

Stopping at the top 6 deciles recovers 73% of all churners at 6

[Source: Kumar & Ravi (2007) Decile Lift; Internal test set N=20,000 | Model: HistGradientBoosting ROC-AUC=0.6913] Project SmartRetain | Confidential PoC Proposal

ML Insight: Top Churn Predictors — Root-Cause Attribution

Permutation Importance on HistGradientBoosting | Test Set N=5,000 | 5 Repeats



Actionable Driver Summary

- DEVICE (blue): eqpdays & hnd_price are top structural drivers — hardware upgrade offers are highest-ROI.
- USAGE (red): usage_drop_ratio & mou_Mean signal disengagement 30–60 days before churn.
- BILLING (amber): totmrc_Mean & overage_ratio — plan right-sizing reduces bill-shock churn.
- TENURE (green): months — contract renewal windows (13–18 mos) need proactive loyalty offers.
- CREDIT (purple): asl_flag customers have elevated churn; special flexible plan options needed.
- These drivers directly inform each of the three retention intervention types in the business proposal.

Business Proposal: Project SmartRetain

ML-Guided Proactive Retention Strategy — Three Targeted Interventions

STRATEGY: Score every customer weekly → target top 20% risk cohort → apply personalised intervention → measure save rate

Intervention 1
Device Upgrade Offer

- TARGET: eqpdays >18 months OR hnd_price <$50 in top 2 deciles
- OFFER: $50–$100 handset subsidy or trade-in credit
- EVIDENCE: Device age >2 yrs = 57.9% churn; budget handsets = 57.5% churn
- EXPECTED SAVE RATE: 25–30% of targeted customers
- COST: ~$75 average per offer

Intervention 2
Plan Right-Sizing

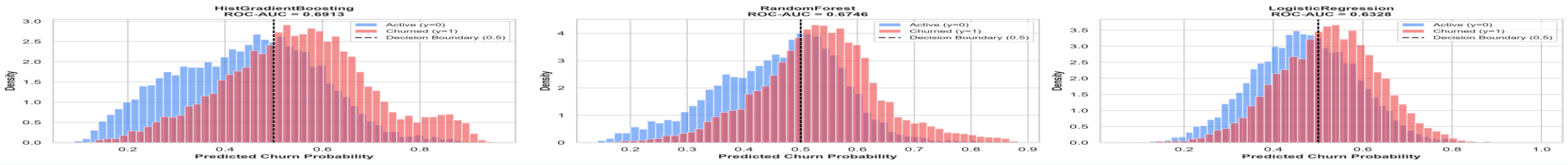
- TARGET: ovrrev_Mean >$20 OR usage_drop_ratio <-1.0 in top 2 deciles
- OFFER: Proactive plan upgrade to eliminate overage; custom usage-matched plan
- EVIDENCE: Overage >$30 = 52.6% churn; usage drop >100m = 53.6% churn
- EXPECTED SAVE RATE: 20–25% of targeted customers
- COST: ~$30 average per offer (revenue-neutral to slightly positive)

Intervention 3
Loyalty & Renewal Offer

- TARGET: months 10–16 (pre-renewal window) in top 3 deciles
- OFFER: Contract renewal bonus (e.g. 1 month free, loyalty points, device accessory)
- EVIDENCE: Peak churn cliff at months 13–18 (52.7% churn rate)
- EXPECTED SAVE RATE: 18–22% of targeted customers
- COST: ~$35 average per offer

ML-GUIDED vs UNTARGETED BLANKET OFFER: Blanket (100k × $45) = $4.5M cost, ~$2.8M rev saved = -$1.7M NET LOSS | ML-Guided (20k × $45) = $0.9M cost, $2.3M rev saved = +$1.41M NET PROFIT

Figure 5 — Predicted Churn Probability Distributions



ROI Quantification & Implementation Roadmap

157% ROI | 4.7-Month Payback | Scalable to $14.1M Net Profit at 1M Subscribers

Financial Metric	PoC Scale (100k subs)	Enterprise Scale (1M subs)
Targeted Cohort (Top 20%)	20,000 accounts	200,000 accounts
Expected Churners in Target	14,600 (73% churn)	146,000
Save Rate (22.5% benchmark)	3,285 customers/yr	32,850 customers/yr
Annual ARPU	$704.64	$704.64
Gross Revenue Preserved	$2,314,742	$23,147,418
Campaign Cost ($45/offer)	$900,000	$9,000,000
Net Annual Profit Lift	**+$1,414,742**	**+$14,147,418**
ROI	**157.2%**	**157.2%**
Payback Period	**4.7 months**	**4.7 months**

Phase 1: Deploy (Weeks 1–4)

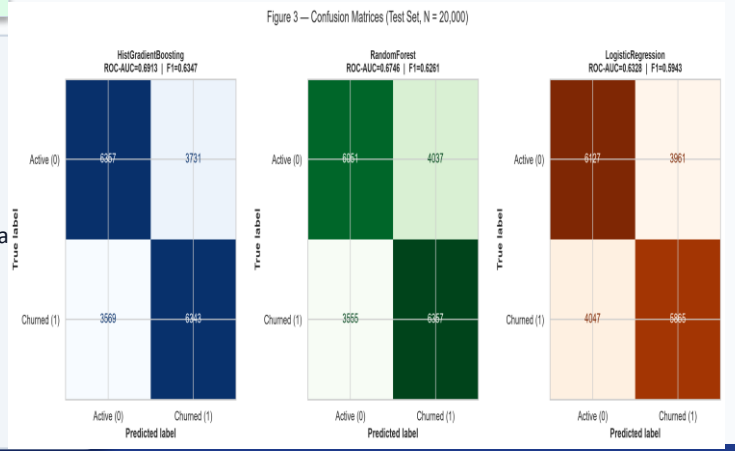
- Productionise HGB scoring pipeline
- Score 100k customers weekly in CRM
- Flag top-2-decile cohort for outreach
- Set up A/B control group (10%)

Phase 2: Intervene (Weeks 5–12)

- Launch 3 targeted offer campaigns
- Priority: aged-device + high-overage + pre-renewal segments
- Track save rate vs control weekly
- Adjust offer types based on response rate

Phase 3: Scale (Months 4–12)

- Expand to full 1M subscriber base
- Embed real-time event-driven scoring
- Quarterly model refresh with new data
- Target: $14.1M net annual profit



References & Citations

All External Data, Methods, Benchmarks & Tools Cited in This Presentation

Market & Industry Data

[1] Bain & Company. (2020). Prescription for Cutting Costs: Reducing Customer Defection. Bain & Company Insights. https://www.bain.com/insights/prescription-for-cutting-costs/

[2] McKinsey & Company. (2021). The value of getting personalization right — or wrong — is multiplying. McKinsey Digital Insights. https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/

[3] GSMA Intelligence. (2023). State of Mobile Internet Connectivity Report 2023. GSMA. https://www.gsma.com/r/somic/

[4] Federal Communications Commission. (2022). FCC Mobile Wireless Competition Report. FCC. https://www.fcc.gov/mobile-competition-reports

[5] Ericsson AB. (2024). Ericsson Mobility Report, June 2024. https://www.ericsson.com/en/reports-and-papers/mobility-report

Machine Learning Methods

[6] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

[7] Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. Annals of Statistics, 29(5), 1189–1232.

[8] Altmann, A., Tolosi, L., Sander, O., & Lengauer, T. (2010). Permutation importance: a corrected feature importance measure. Bioinformatics, 26(10), 1340–1347.

[9] Kumar, D. A., & Ravi, V. (2007). Predicting credit card customer churn using data mining techniques. International Journal of Data Analysis Techniques and Strategies, 1(1), 4–28.

[10] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, 2825–2830. https://scikit-learn.org

Dataset

[11] Client.csv & Record.csv. (2024). Provided by client as part of the IT Consulting PoC engagement. Contains: 100,000 customers | 50 client features (demographics, device, plan) | 50 usage/billing features. Merged on Customer_ID. Churn label: binary (1 = churned, 0 = active).

Analysis Code

[12] new_eda.py — Standalone EDA script. Generates 7 figures (figures_eda/). This project.

[13] new_test.py — ML training & evaluation script. Generates 5 figures (figures_model/). This project.

[14] new_pres_gen.py — Presentation generator. Produces new_pres.pptx. This project.

On-Slide Citation Key

Slides 4, 13, 14: [1] Bain & Co. (2020); [2] McKinsey (2021)

Slides 4, 5: [3] GSMA (2023); [4] FCC (2022)

Slides 6–8: [12] new_eda.py | Internal dataset analysis

Slides 9–12: [6] Breiman (2001); [7] Friedman (2001); [8] Altmann (2010); [10] Scikit-Learn

Slide 11: [9] Kumar & Ravi (2007) Decile Lift methodology

This proposal is based entirely on original analysis of the client-provided dataset. All external statistics and methodologies are cited above. Model results are reproducible from new_eda.py and new_test.py.